i-CAPE iSO iTSE

Innovative Science Olympiad

2025 – 2026

SET - A

Grade 9

Total Questions: 50 | Time: 1.5 hr. **Total Marks: 100**

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. A farmer practices a cropping pattern where he grows Soybeans and Maize in the same field in a 1:2 row ratio. He notices that the incidence of pests is lower compared to when he grew them separately. **Identify the practice and the reason for the yield stability.**

(A) Crop Rotation; it replenishes soil nitrogen.

(B) Mixed Cropping; it ensures insurance against crop failure.

(C) Intercropping; it prevents pests from spreading to all plants of one species.

(D) Intensive Farming; it uses high chemical inputs for maximum growth.

2. A scuba tank is filled with a gas mixture containing only nitrogen (N_2) and oxygen (O_2). At STP, the total volume of gas in the tank is 33.6 L and the mass of oxygen present is 16 g. Calculate the mass of nitrogen present in the tank.

(A) 14 g (B) 21 g

(C) 28 g (D) 42 g

3. A scientist studies two hydrogen-like ions, **Ion A** and **Ion B**.

For Ion A, the electron is moving in the second orbit ($n_A = 2$) with atomic number $Z_A = 2$.

Ion B is a hydrogen-like ion with atomic number $Z_B = 4$. The electron in Ion B is moving in an orbit such that its velocity is exactly equal to that of the electron in Ion A. Determine the ratio of the principal quantum numbers of Ion B to Ion A.

(A) 1 : 2 (B) 2 : 1

(C) 1 : 4 (D) 4 : 1

4. A chemistry student prepares 200 g of a sugar solution of concentration 12% (w/w). She then adds 100 g of pure water to it. **What is the new weight by weight percentage of sugar in the resulting solution?**

(A) 6% (B) 8%

(C) 9% (D) 12%

5. A scientist treats a sample of plant tissue with a chemical that dissolves pectin, the primary component of the middle lamella. Upon observation under a high-resolution microscope, **what will be the immediate physical effect on the tissue structure?**

(A) The individual cells will shrink and undergo plasmolysis.

(B) The cells will lose their internal genetic material as the nucleus dissolves.
(C) The individual cells will separate from one another, as the "glue" holding them together is gone.
(D) The xylem vessels will double their rate of water conduction to compensate for the loss.

6. A narrow beam of light is passed through three mixtures:
I: Salt in water
II: Milk in water
III: Mud in water
Only mixture II shows a visible path of light, and its particles do not settle on standing.
What does mixture II represent?
(A) True solution (B) Colloid
(C) Suspension (D) Pure substance

7. An atom with atomic number 12 is most likely to combine chemically with an atom whose atomic number is ______.
(A) 9 (B) 10
(C) 18 (D) 20

8. A specific cell organelle is described as having its own DNA and ribosomes, and it is involved in the conversion of light energy into chemical energy. If this organelle is removed from a cell, **what is the most likely consequence?**
(A) The cell will be unable to break down complex organic substances into simpler ones.
(B) The cell will lose its ability to undergo aerobic respiration.
(C) The cell will be unable to synthesize its own food using sunlight.
(D) The cell will immediately lose its structural integrity and collapse.

9. A sealed container of volume 500 cm^3 has a mass of 600 g. If this container is dropped into a tank of water, will it float or sink? Also, what is the magnitude of the net force acting on it while it is fully submerged? (Density of water = 1 g/cm^3, g = 10 m/s^2)
(A) Floats, Net Force = 0 N
(B) Sinks, Net Force = 1 N downwards
(C) Sinks, Net Force = 6 N downwards
(D) Floats, Net Force = 5 N upwards

10. Match the columns and select the correct option.

	Column I (Transition elements)		Column II (Electronic configuration)
P.	Chromium (Cr)	**(i)**	[Ar] $3d^{10}\,4s^1$
Q.	Copper (Cu)	**(ii)**	[Ar] $3d^5\,4s^1$
R.	Iron (Fe)	**(iii)**	[Ar] $3d^6\,4s^2$
S.	Zinc (Zn)	**(iv)**	[Ar] $3d^{10}\,4s^2$

(A) P → (ii), Q → (i), R → (iii), S → (iv)
(B) P → (iii), Q → (ii), R → (iv), S → (i)
(C) P → (ii), Q → (iv), R → (i), S → (iii)
(D) P → (i), Q → (ii), R → (iii), S → (iv)

11. **Fill in the cells in the given table by selecting the correct option.**

S. No.	Dispersed Phase	Dispersion Medium	Type of Colloid	Example
1.	Solid	(i)	Sol	Starch in water
2.	(iii)	Solid	Solid foam	Sponge
3.	Liquid	Liquid	(ii)	Milk
4.	Liquid	Gas	Aerosol	(iv)

	(i)	(ii)	(iii)	(iv)
(A)	Liquid	Suspension	Solid	Mist
(B)	Liquid	Emulsion	Gas	Fog
(C)	Gas	Sol	Liquid	Smoke
(D)	Solid	Emulsion	Gas	Smoke

12. Four situations are given below:

I. Water kept in an open dish at room temperature
II. Water heated in a covered beaker until bubbles appear
III. Perfume spilled on the floor
IV. Water kept in a closed bottle at room temperature

Which of the following statements is correct regarding evaporation and boiling?

(A) Situation II represents evaporation because it occurs only at the surface of water.
(B) Situation III represents boiling because the liquid changes into vapour rapidly.
(C) Situation I represents evaporation because it occurs at all temperatures and only at the surface of the liquid.
(D) Situation IV represents boiling because the vapour cannot escape from the bottle.

13. A passenger is sitting in a high-speed train moving at a constant velocity of 120 km/h. He tosses a coin vertically upwards. To his surprise, the coin falls exactly back into his hand. Which of the following statements best explains this observation in terms of the frame of reference?

(A) The coin gains additional horizontal force from the person's hand during the toss.
(B) In the passenger's frame of reference, the coin has only vertical motion, while for an observer on the platform, the coin follows a parabolic path.
(C) The train's speed is so high that gravity acts differently on the coin inside the compartment.
(D) The air resistance inside the train pushes the coin forward to match the train's speed.

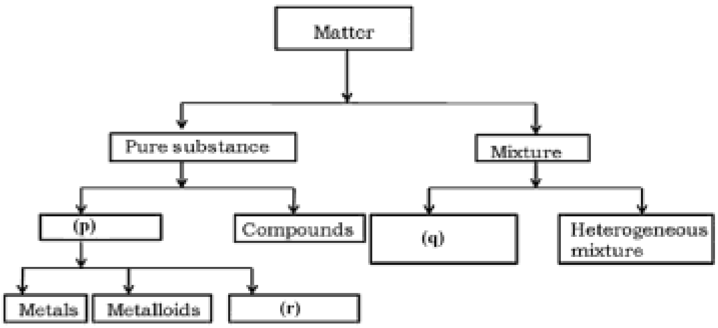
14. A bullet is fired into a wooden plank with a velocity of 100 m/s. It penetrates 5 cm before coming to rest. If the thickness of the plank was only 3 cm, with what velocity would the bullet have emerged from the other side, assuming the same retardation?

(A) 60 m/s (B) 63.2 m/s
(C) 40 m/s (D) 50 m/s

15. Select the **incorrect statement.**

(A) A solution of water and sugar is called syrup.
(B) A solution of water and salt is called brene.
(C) A solution of carbon dioxide in water is called seltzer.
(D) A solution of ammonia gas in water is called ammonia water.

16. Fill in the boxes by selecting the correct option.



	(p)	(q)	(r)
(A)	Elements	Homogeneous mixture	Non-metals
(B)	Compounds	Homogeneous mixture	Metals
(C)	Elements	Solution	Metalloids
(D)	Mixture	Homogeneous mixture	Non-metals

17. A laboratory technician prepares two different solutions:

Solution I: 30 mL of ethanol is mixed with water to make 200 mL of solution.

Solution II: 12 g of common salt is dissolved in water to make 100 mL of solution.

Which of the following correctly represents the percentage concentration and the method of expression for both solutions?

(A) Solution I: 15% (v/v), Solution II: 12% (w/v)

(B) Solution I: 30% (v/v), Solution II: 12% (w/w)

(C) Solution I: 15% (w/v), Solution II: 12% (v/v)

(D) Solution I: 15% (v/v), Solution II: 12% (w/w)

18. Calculate the energy required to remove an electron from the energy level n = 3 of a hydrogen-like ion with atomic number $Z = 18$.

(A) 163.2 eV (B) 244.8 eV
(C) 489.6 eV (D) 979.2 eV

19. Two particles **R** and **S** have the composition as shown in the table:

Species	Number of protons	Number of neutrons	Number of electrons
R	11	12	10
S	9	10	10

The particles **R** and **S** are:

(A) Isoelectronic species
(B) Isotopes of the same element
(C) Isobars
(D) Isoelectronic and isotopic species

20. **Assertion (A):** No two electrons present in the same atom can have identical values of all four quantum numbers.

Reason (R): Electrons in the same orbital must have the same spin quantum number.

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion
(B) Both Assertion and Reason are true, but Reasons is not the correct explanation of Assertion
(C) Assertion is true, but Reason is false
(D) Assertion is false, but Reason is true

21. Adipose tissue acts as an insulator. Why is this tissue particularly thick in animals living in extremely cold climates, like polar bears?

(A) It provides the energy required to hunt in the snow.
(B) The storage of fat globules acts as a thermal barrier, preventing the loss of body heat.
(C) It makes the animal lighter, allowing it to walk on thin ice.
(D) It absorbs sunlight and converts it into heat for the internal organs.

22. A scientist is working on "Genetic Manipulation" to create a new variety of rice. **What is the ultimate goal of this specific process compared to traditional breeding?**

(A) To increase the size of the seeds only.
(B) To incorporate a specific desirable gene from another organism into the plant's DNA.
(C) To ensure the plant grows without any water.
(D) To make the rice taste like wheat.

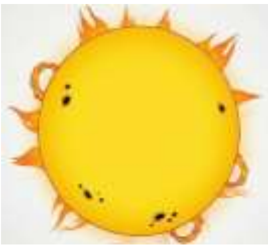
23. Substance R at 40 °C has neither definite shape nor definite volume. At 5°C, substance R has definite volume but no definite shape. Which of the following could be the condensation point and freezing point of substance R?

	Condensation point	**Freezing point**
(A)	45 °C	0°C
(B)	30 °C	-10 °C
(C)	60 °C	-20 °C
(D)	50 °C	20 °C

24. If the Earth suddenly shrank to half its current radius while keeping its mass constant, **what would happen to the weight of an object on its surface?**

(A) The weight would become half.
(B) The weight would remain the same.
(C) The weight would become four times greater.
(D) The weight would become double.

25. Which state of matter is represented in the below diagram?



Sun

(A) Solid (B) Liquid
(C) Gas (D) Plasma

26. An astronaut on a mystery planet drops a 2 kg object from a height of 10 m, and it hits the ground in 2 seconds. If the radius of this planet is the same as Earth's, what is the mass of this planet compared to Earth's mass (M_e)?

(A) $0.5M_e$ (B) $2M_e$
(C) $0.25M_e$ (D) M_e

27. A cell's ability to exchange materials with its environment is limited by its surface-area-to-volume ratio. **Why are most cells microscopic in size?**

(A) Smaller cells have a larger surface area relative to their volume, allowing for efficient nutrient intake and waste removal.
(B) Larger cells require more gravity to stay stable, which would crush the delicate organelles.
(C) The nucleus cannot produce enough DNA to fill a large cell.
(D) Smaller cells move faster, allowing the organism to react to danger more quickly.

28. Suppose a drug is administered that inhibits the formation of myelin sheaths in the nervous system. **What would be the most likely analytical effect on nerve impulse transmission?**

(A) The impulses would stop entirely and the neuron would die.
(B) The impulses would travel much faster because they don't have to go through the sheath.
(C) The impulses would travel significantly slower and might "leak" or dissipate.
(D) The dendrites would turn into axons.

29. An archer pulls the string of her bow back by 0.5 m with an average force of 200 N. She holds it perfectly still for 3 seconds while aiming at the target, then releases the 0.02 kg arrow. **Which of the following is scientifically correct regarding this scenario?**
(A) The archer does 600 J of work while holding the bow still.
(B) The potential energy stored in the bow is 100 J, which is then converted into the kinetic energy of the arrow.
(C) No work is done by the archer at any point because the arrow eventually leaves the bow.
(D) The power expended by the archer while aiming is 200 W.

30. **Assertion (A):** Broilers are fed with vitamin-rich supplementary feed for good growth rate and better feed efficiency.
Reason (R): Broilers are raised primarily for egg production.
(A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

31. A small hydroelectric power plant is built near a waterfall where 2,000 kg of water falls every second from a height of 30 meters. The turbine at the bottom is only 60% efficient in converting the falling water's energy into electricity. **What is the actual electrical power output of this plant?** (Take g = 10 m/s²)
(A) 600 kW (B) 360 kW
(C) 120 kW (D) 36,00 W

32. A laboratory technician has three sealed containers:
Container X has 11 g of CO_2
Container Y has 5.8 g of NaCl
Container Z has 3.01×10^{23} molecules of H_2O
Using the concept of mole, determine which container contains exactly 0.5 mole of substance.
(A) Only container X
(B) Only container Y
(C) Only container Z
(D) Containers X and Y

33. Valency of Sulphur in SO_3 and copper in cupric chloride is _____.
(A) 3, 2 (B) 6, 2
(C) 5, 1 (D) 4, 4

34. A driver of a car travelling at 72 km/h sees a child on the road 50 m ahead. He takes 0.5 seconds to react (reaction time) and then applies brakes, causing a uniform retardation of 5 m/s². Does he hit the child? If not, what is the final distance between the car and the child?
(A) Yes, he hits the child.
(B) No, the car stops exactly at 20 m.
(C) No, the car stops 10 m before the child.
(D) No, the car stops 0 m before the child.

35. A stone is dropped from the top of a tower 100 m high. At the same time, another stone is thrown vertically upwards from the ground with a velocity of 25 m/s. At what time and height from the ground will the two stones pass each other? (Take $g = 10$ m/s^2)
(A) 4 s, 80 m (B) 2 s, 20 m
(C) 4 s, 20 m (D) 5 s, 25 m

36. Some elements are named in honour of places. Which of the following elements is named after a country?
(A) Polonium (B) Helium
(C) Silicon (D) Mercury

37. A motor pumps 500 kg of water up to an overhead tank at a height of 10 m in 50 seconds. If only 80% of the motor's power is effectively used to lift the water, what is the total power rating of the motor? **(Take g = 10 m/s²)**

(A) 1000 W (B) 1250 W
(C) 800 W (D) 1500 W

38. What happens to the volume of the solution when a small amount of sugar is dissolved in it?

(A) Volume will increase
(B) Volume will decrease
(C) Volume first increases then decreases
(D) No change in volume

39. A robotic space probe of mass 1,000 kg is traveling through deep space (far from any planets) at a constant velocity of 5,000 m/s. The mission control decides to test the small thrusters, which apply a constant force of 500 N in the direction of motion for a distance of 2 km. What is the final kinetic energy of the probe after the thrusters have been fired for those 2 km?

(A) 12.5×10^9 J (B) 1.25×10^7 J
(C) 1250.1×10^9 J (D) 1.2504×10^7 J

40. In a hydroelectric power plant, water falls from a height of 100 m to turn a turbine. If 2,000 kg of water falls every second, **what is the ideal electrical power generated?** (Assume 100% efficiency and g = 10 m/s²)

(A) 2 MW (B) 200 kW
(C) 200 W (D) 20 MW

41. A Delhi Metro train is approaching a station at 20 m/s. The automated braking system is programmed to bring the train to a halt over a distance of 200 m with uniform retardation. However, due to a technical glitch, the brakes only provide half the required retardation for the first 100 m. If the system fixes itself after 100 m and applies the original intended retardation, **where will the train stop?**

(A) Exactly at the station.
(B) 50 m past the station.
(C) 100 m past the station.
(D) 25 m before the station.

42. A 50 g bullet is fired from a 4 kg rifle with an initial velocity of 35 m/s. The rifle recoils and comes to rest after sliding 0.5 m on a rough surface. **What is the recoil velocity of the rifle and the acceleration it experiences while sliding?**

(A) $v = 0.4$ m/s, $a = -0.19$ m/s^2
(B) $v = 0.875$ m/s, $a = -0.76$ m/s^2
(C) $v = 1.25$ m/s, $a = -1.56$ m/s^2
(D) $v = 0.08$ m/s, $a = -0.12$ m/s^2

43. **Assertion (A):** Ciliated columnar epithelium is found in the lining of the respiratory tract.
Reason (R): The cilia move rhythmically to push mucus and trapped dust forward to clear the passage.

(A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

44. How does the inertia of an object change if its mass is tripled?

(A) The inertia remains the same because inertia only depends on speed.
(B) The inertia becomes one-third of the original.
(C) The inertia becomes three times the original.
(D) The inertia increases by a factor of nine (3^2).

45. A fish farmer has a pond where he grows only **'Catla'.** He notices that the food at the bottom of the pond is remaining uneaten, leading to water pollution, while the surface food is being exhausted rapidly. To optimize the pond's **"carrying capacity," which species should he introduce?**

(A) More Catla to eat the surface food faster.
(B) Rohu to feed in the middle zone and Mrigals/Common Carps to feed at the bottom.
(C) Sharks to keep the fish population healthy.
(D) Aquatic weeds to absorb the excess nutrients.

46. A motorcar of mass 1200 kg is moving along a straight line with a uniform velocity of 90 km/h. Its velocity is slowed down to 18 km/h in 4 seconds by an external unbalanced force. **Calculate the acceleration and the change in momentum.**

(A) $a = -5$ m/s^2, $\Delta p = -24{,}000$ kg.m/s
(B) $a = -18$ m/s^2, $\Delta p = -86{,}400$ kg.m/s
(C) $a = 5$ m/s^2, $\Delta p = 24{,}000$ kg.m/s
(D) $a = -2.5$ m/s^2, $\Delta p = -12{,}000$ kg.m/s

47. A constant force acts on an object of mass 10 kg for a duration of 2 seconds. It increases the object's velocity from 3 m/s to 7 m/s. Now, if the same force were applied for a total of 5 seconds, **what would be the final velocity of the object?**

(A)10 m/s (B) 13 m/s
(C) 15 m/s (D) 17 m/s

48. An investigator is studying three unknown cell samples (A, B, and C).

- Cell A has a cell wall, but no nuclear membrane.
- Cell B has a cell wall, a nuclear membrane, and large vacuoles.
- Cell C has no cell wall, but possesses centrioles and multiple small vacuoles.

Which of the following correctly identifies these cells

(A) A: Plant Cell; B: Bacterial Cell; C: Animal Cell
(B) A: Bacterial Cell; B: Plant Cell; C: Animal Cell
(C) A: Animal Cell; B: Bacterial Cell; C: Plant Cell
(D) A: Bacterial Cell; B: Animal Cell; C: Plant Cell

49. A rectangular wooden block of dimensions 10 cm × 20 cm × 50 cm is placed on a table. If the mass of the block is 5 kg, calculate the maximum pressure it can exert on the table. **(Take g = 10 m/s²)**

(A) 500 Pa (B) 2500 Pa
(C) 1000 Pa (D) 5000 Pa

50. A ball is thrown vertically upwards and reaches a maximum height of 20 m. Calculate the velocity with which it was thrown, and determine the pressure exerted by this ball on the ground if it has a mass of 500 g and comes to rest on an area of 0.005 m² after falling back. **(Take g = 10 m/s²)**

(A) $u = 20$ m/s,Pressure = 1000 Pa
(B) $u = 10$ m/s,Pressure = 500 Pa
(C) $u = 20$ m/s,Pressure = 5000 Pa
(D) $u = 4\ 0$ m/s,Pressure = 2500 Pa

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